

Anesthesia for ENT Surgery MCQ – Chapter 34

Source: Short Textbook of Anesthesia – Ajay Yadav

Section 8: Subspecialty Anesthetic Management (Chapter 34)

50 Questions • 30 Minutes

Instructions:

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. Why is ENT anesthesia considered one of the most difficult?

- A. Due to patient age
- B. Due to lengthy procedures
- C. Because airway is often difficult, surgeon and anesthetist share the common field, and complications are related to airway
- D. Due to bleeding only

Ans: C

Q2. What does panendoscopy involve?

- A. Only esophagoscopy
- B. Laryngoscopy, bronchoscopy, and esophagoscopy (also called triple endoscopy)
- C. Only laryngoscopy
- D. Only bronchoscopy

Ans: B

Q3. What happens to the glottic opening due to laryngeal growth in panendoscopy patients?

- A. It remains normal
- B. It may be reduced to the extent that even small tube insertion becomes impossible
- C. It becomes wider
- D. No change occurs

Ans: B

Q4. What is the incidence of myocardial ischemia and arrhythmias due to laryngeal manipulation?

- A. 10-15%
- B. Less than 1%
- C. 2-4%
- D. More than 20%

Ans: C

Q5. What must be done during preoperative evaluation for panendoscopy patients?

- A. Only chest X-ray
- B. Assessment of airway by preoperative endoscopy or indirect laryngoscopy is must
- C. Only blood tests
- D. Only ECG

Ans: B

Q6. Why are IV lignocaine, IV fentanyl and topical anesthesia of the larynx important for panendoscopy?

- A. To reduce bleeding
- B. To prevent laryngeal reflexes
- C. To reduce secretions
- D. To reduce postoperative pain

Ans: B

Q7. What are MLS tubes? A. Double lumen tubes B. Tracheostomy tubes C. Adult size ETT without cuff D. Pediatric size tubes with cuff available in 5, 5.5 and 6 mm size	Ans: D
Q8. In what percentage of patients is it possible to get surgery done with MLS tube? A. 50% B. 25% C. >95% D. 75%	Ans: C
Q9. What is the disadvantage of MLS tubes? A. High ventilatory pressures are required to overcome the narrow size of tube B. They cause more bleeding C. They cannot be used in adults D. They are expensive	Ans: A
Q10. In High Frequency Positive Pressure Ventilation (HFPPV), what is the frequency of breaths delivered? A. 20-40 breaths/min B. 10-20 breaths/min C. 60-100 breaths/min D. 120-150 breaths/min	Ans: C
Q11. What is the diameter of the catheter used in HFPPV? A. 3.5-4.0 mm B. 1.0-1.5 mm C. 5.0-6.0 mm D. 2.0-2.5 mm	Ans: A
Q12. In jet ventilation, where is the tip of the jet kept? A. Within the laryngoscope to avoid direct barotrauma to trachea B. Beyond the vocal cords C. In the trachea directly D. In the pharynx	Ans: A
Q13. What is the apnea technique in ventilation for panendoscopy? A. Mask ventilation only B. High flow nasal oxygen C. Intubation is done, adequate ventilation given, tube removed for 1-2 minutes for surgery, then reintubated if needed D. Patient breathes spontaneously	Ans: C

Q14. How is anesthesia maintained during panendoscopy? A. With inhalational agent and repeated doses of succinylcholine for small procedures or nondepolarizing agent for prolonged surgery B. Only regional anesthesia C. Only IV agents D. Only propofol infusion	Ans: A
Q15. Why should nitrous oxide be avoided in laryngeal surgeries? A. It is not potent enough B. It causes bleeding C. It causes laryngospasm D. It can increase postsurgical edema by causing expansion of the cuff of endotracheal tube	Ans: D
Q16. When is Heliox administration strongly considered in ENT? A. In patients with stridor B. In all ENT surgeries C. Only in pediatric patients D. Only in bronchoscopy	Ans: A
Q17. What is the most important complication of laser surgery? A. Infection B. Fires C. Bleeding D. Nerve injury	Ans: B
Q18. What FiO2 should be used during laser surgery to prevent fire? A. Less than 30% B. 50-60% C. 40-50% D. 100%	Ans: A
Q19. Why should N2O be avoided during laser surgery? A. It is not potent B. It can support fire C. It causes nausea D. It causes laryngospasm	Ans: B
Q20. What special kind of endotracheal tube is used for laser surgery? A. Laser resistant endotracheal tube B. Double lumen tube C. Uncuffed tube D. Reinforced tube	Ans: A

Q21. What should the tube cuff be filled with during laser surgery instead of air? A. Helium B. Saline C. Oxygen D. Nitrous oxide	Ans: B
Q22. What is the FIRST step if fire occurs during laser surgery? A. Call for help B. Stop oxygen C. Immediately remove tube and submerge in water D. Flush pharynx with saline	Ans: C
Q23. What is the second step in the fire protocol after removing the tube? A. Flush with saline B. Give steroids C. Reintubate D. Immediately stop oxygen (and nitrous oxide) and start ventilation with air by mask	Ans: D
Q24. What volume of cold saline should be used to flush the pharynx after airway fire? A. 10-20 mL B. 50-60 mL C. 100-200 mL D. 500 mL	Ans: B
Q25. When may tracheostomy be required after an airway fire? A. Always B. Never needed C. Only in children D. If damage is severe enough not to allow intubation	Ans: D
Q26. Under what type of anesthesia is flexible bronchoscopy for diagnostic biopsies usually performed? A. Regional anesthesia B. Topical anesthesia C. Spinal anesthesia D. General anesthesia	Ans: B
Q27. What is rigid bronchoscopy usually done for? A. Foreign body removal under general anesthesia B. Lavage only C. Diagnostic biopsies D. Stent placement only	Ans: A

Q28. What are the ventilation methods for rigid bronchoscopy? A. High frequency jet ventilation, insufflation, and apnea technique B. Only spontaneous breathing C. Only IPPV D. Only mask ventilation	Ans: A
Q29. What is the majority of patients undergoing adenotonsillectomy? A. Pediatric B. Adolescents C. Adults D. Elderly	Ans: A
Q30. If there are no adenoids, what route should be preferred for nasal intubation? A. Nasal is always preferred B. Only oral RAE tube C. Nasal intubation is always safe D. Nasal intubation should preferably be avoided due to increased chances of bleeding	Ans: D
Q31. What type of endotracheal tubes are most frequently used for tonsillectomy? A. Nasal RAE tubes B. Oral Ring-Adair-Elwyn (RAE) preformed tubes C. Standard ETT D. Reinforced tubes	Ans: B
Q32. Why should the throat be packed during tonsillectomy? A. To prevent aspiration of blood B. To reduce swelling C. To improve ventilation D. To keep the tube in place	Ans: A
Q33. In what position should tonsillectomy patients be nursed postoperatively? A. Sitting upright B. Prone position C. Tonsillar position - on one side (preferably left lateral) with head low D. Supine with head elevated	Ans: C
Q34. How should a patient with post-tonsillectomy bleeding requiring re-exploration be managed? A. As a full stomach patient (patient has swallowed blood causing significant aspiration risk) B. Only with sedation C. Only under local anesthesia D. As a routine case	Ans: A

Q35. How should peritonsillar abscess and Ludwig angina ideally be drained? A. Under regional block B. Under local anesthesia (as intubation may be impossible due to trismus and aspiration risk) C. Under general anesthesia always D. Under IV sedation	Ans: B
Q36. If GA is required for peritonsillar abscess, what should be done before induction? A. Rapid sequence induction B. Blind nasal intubation C. Awake nasal intubation with fiberoptic bronchoscopy D. LMA insertion	Ans: C
Q37. How many times more soluble is nitrous oxide compared to nitrogen? A. 10 times B. 35 times C. 20 times D. 50 times	Ans: B
Q38. By how much can nitrous oxide increase middle ear pressure if Eustachian tube is blocked? A. 375 mm H₂O within half an hour B. 200 mm H₂O in 2 hours C. 50 mm H₂O in 15 minutes D. 100 mm H₂O in 1 hour	Ans: A
Q39. In which ear conditions should nitrous oxide NOT be used? A. Only otitis media B. Patients with blocked Eustachian tubes (upper respiratory tract infection, adenoids, sinusitis and otitis media) C. All ear surgeries D. Only sinusitis	Ans: B
Q40. Why is nitrous oxide contraindicated for tympanoplasties? A. Increase in middle ear pressure can displace the graft B. It causes nausea C. It causes infection D. It causes bleeding	Ans: A
Q41. When is increased middle ear pressure from N₂O beneficial in ear surgery? A. Never B. In mastoidectomy C. In all tympanoplasties D. When surgeon is using 'inlay graft' (helps hold the graft)	Ans: D

Q42. What is very high in incidence after ear surgeries requiring prophylaxis? A. Pain B. Postoperative nausea and vomiting C. Bleeding D. Infection	Ans: B
Q43. What is the Samter triad that must be ruled out in nasal surgery patients? A. Deviated septum, sinusitis, rhinitis B. Hypertension, diabetes, obesity C. Bleeding, infection, pain D. Nasal polyps, asthma, and sensitivity to aspirin/NSAIDs	Ans: D
Q44. Why is throat pack a must in nasal surgery? A. To improve ventilation B. To reduce swelling C. To prevent aspiration of blood D. To stabilize the tube	Ans: C
Q45. Why must nasal surgery patients be counseled to breathe from mouth preoperatively? A. For better oxygenation B. To reduce anxiety C. Because they can become very restless in postoperative period after nasal packing if not trained D. To improve lung function	Ans: C
Q46. Why may muscle relaxants be deterred in parotid surgery? A. Due to patient positioning B. Due to risk of bleeding C. Due to risk of aspiration D. Because continuous monitoring of facial nerve function is needed	Ans: D
Q47. Why is obstructive sleep apnea surgery very challenging for anesthetists? A. Due to bleeding only B. Due to obesity only C. All concerns related to obese patients especially difficult airway management D. Due to long duration only	Ans: C
Q48. What does limited mouth opening in TMJ surgeries mandate? A. Oral intubation under GA B. LMA use C. Tracheostomy D. Awake nasal intubation	Ans: D

Q49. Are laser resistant endotracheal tubes 100% protective against laser fires?

- A. No, they are not 100% protective against laser fires
- B. Yes, completely protective
- C. Only metal tubes are protective
- D. Only silicone tubes are protective

Ans: A

Q50. What laryngeal complication can stimulation during panendoscopy induce?

- A. Vocal cord paralysis
- B. Laryngeal edema
- C. Subglottic stenosis
- D. Laryngospasm

Ans: D

Answer Key & Explanations

Q1. Answer: C — Because airway is often difficult, surgeon and anesthetist share the common field, and complications are related to airway

ENT anesthesia is one of the most difficult because quite often airway is difficult, surgeon and anesthetist share the common field and complications are related to airway.

Q2. Answer: B — Laryngoscopy, bronchoscopy, and esophagoscopy (also called triple endoscopy)

Panendoscopy involves laryngoscopy, bronchoscopy, and esophagoscopy. It is also called as triple endoscopy.

Q3. Answer: B — It may be reduced to the extent that even small tube insertion becomes impossible

Due to the growth, the glottic opening may be reduced and sometimes to an extent that even insertion of the small size tube becomes impossible.

Q4. Answer: C — 2-4%

Laryngeal manipulation causes sympathetic stimulation. Incidence of myocardial ischemia and arrhythmias is significant (2-4%).

Q5. Answer: B — Assessment of airway by preoperative endoscopy or indirect laryngoscopy is must

During preoperative evaluation, assessment of airway by preoperative endoscopy or indirect laryngoscopy is must.

Q6. Answer: B — To prevent laryngeal reflexes

Intravenous (IV) lignocaine, IV fentanyl and topical anesthesia of the larynx are very important to prevent laryngeal reflexes.

Q7. Answer: D — Pediatric size tubes with cuff available in 5, 5.5 and 6 mm size

Microlaryngeal surgery (MLS) tubes are pediatric size tubes with cuff. They are available in 5, 5.5 and 6 mm size.

Q8. Answer: C — >95%

In the majority (>95%) of the patients it is possible to get the surgery done with MLS tube.

Q9. Answer: A — High ventilatory pressures are required to overcome the narrow size of tube

The disadvantage is that high ventilatory pressures are required to overcome the narrow size of tube.

Q10. Answer: C — 60-100 breaths/min

In HFPPV, small tidal volumes are delivered at high frequency (60-100 breaths/min) at a high pressure (60 psi) through a stiff 3.5-4.0 mm catheter passed through the vocal cord.

Q11. Answer: A — 3.5-4.0 mm

A stiff 3.5-4.0 mm catheter is passed through the vocal cord for HFPPV.

Q12. Answer: A — Within the laryngoscope to avoid direct barotrauma to trachea

The tip of the jet is kept within the laryngoscope to avoid direct barotrauma to the trachea and gas is delivered at high pressure (50-60 psi).

Q13. Answer: C — Intubation is done, adequate ventilation given, tube removed for 1-2 minutes for surgery, then reintubated if needed

In apnea technique intubation is done and after adequate ventilation tube is removed for short period (1-2 minutes or till oxygen saturation starts falling). In this period surgeon performs the procedure.

Q14. Answer: A — With inhalational agent and repeated doses of succinylcholine for small procedures or nondepolarizing agent for prolonged surgery

Anesthesia is maintained with inhalational agent and repeated doses of succinylcholine for small procedures or nondepolarizing agent for prolonged surgery.

Q15. Answer: D — It can increase postsurgical edema by causing expansion of the cuff of endotracheal tube

Nitrous oxide should be avoided because it can increase the postsurgical edema by causing expansion of the cuff of endotracheal tube.

Q16. Answer: A — In patients with stridor

Administration of Heliox is strongly considered in patients with stridor.

Q17. Answer: B — Fires

The most important complication of laser surgery is fires, therefore the most important aim should be to prevent fire.

Q18. Answer: A — Less than 30%

Use low FiO₂ (<30%) to prevent fire during laser surgery.

Q19. Answer: B — It can support fire

Avoid N₂O as it can support fire.

Q20. Answer: A — Laser resistant endotracheal tube

Use special kind of laser resistant endotracheal tube.

Q21. Answer: B — Saline

Fill the tube cuff with saline (instead of air) during laser surgery.

Q22. Answer: C — Immediately remove tube and submerge in water

If fire occurs: immediately remove tube and submerge in water.

Q23. Answer: D — Immediately stop oxygen (and nitrous oxide) and start ventilation with air by mask

Immediately stop oxygen (and nitrous oxide) and start ventilation of the patient with air by mask.

Q24. Answer: B — 50-60 mL

Flush pharynx with 50-60 mL of cold saline.

Q25. Answer: D — If damage is severe enough not to allow intubation

Tracheostomy may be required if damage is severe enough not to allow intubation.

Q26. Answer: B — Topical anesthesia

Flexible bronchoscopy, usually done for diagnostic biopsies, can be performed under topical anesthesia.

Q27. Answer: A — Foreign body removal under general anesthesia

Rigid bronchoscopy, usually done for foreign body removal is performed under general anesthesia.

Q28. Answer: A — High frequency jet ventilation, insufflation, and apnea technique

Ventilation during rigid bronchoscopy is achieved with: high frequency jet ventilation through the oxygen port, insufflation, and apnea technique.

Q29. Answer: A — Pediatric

Majority of the patients for adenotonsillectomy/tonsillectomy are pediatric.

Q30. Answer: D — Nasal intubation should preferably be avoided due to increased chances of bleeding

If there are no adenoids nasal intubation can be done, however due to increased chances of bleeding nasal intubation should preferably be avoided.

Q31. Answer: B — Oral Ring-Adair-Elwyn (RAE) preformed tubes

Oral Ring-Adair-Elwyn (RAE) preformed tubes are most frequently used.

Q32. Answer: A — To prevent aspiration of blood

Throat should be packed to prevent aspiration.

Q33. Answer: C — Tonsillar position - on one side (preferably left lateral) with head low

These patients are nursed in tonsillar position, i.e. on one side (preferably left lateral) with head low to prevent aspiration and laryngospasm.

Q34. Answer: A — As a full stomach patient (patient has swallowed blood causing significant aspiration risk)

Such patient should be resuscitated and managed like full stomach patient considering that patient has swallowed enough blood which can cause significant aspiration.

Q35. Answer: B — Under local anesthesia (as intubation may be impossible due to trismus and aspiration risk)

General anesthesia should be avoided as intubation at times may be impossible due to trismus and patient can aspirate the pus. Therefore, the abscesses should be drained under local anesthesia.

Q36. Answer: C — Awake nasal intubation with fiberoptic bronchoscopy

If GA is required than awake nasal intubation with fiberoptic bronchoscopy should be achieved before induction of anesthesia.

Q37. Answer: B — 35 times

Nitrous oxide is 35 times more soluble than nitrogen.

Q38. Answer: A — 375 mm H2O within half an hour

If the Eustachian tube is blocked, nitrous oxide can increase the middle ear pressure to 375 mm H2O within half an hour.

Q39. Answer: B — Patients with blocked Eustachian tubes (upper respiratory tract infection, adenoids, sinusitis and otitis media)

Nitrous oxide should not be used in patients with blocked Eustachian tubes (upper respiratory tract infection, adenoids, sinusitis and otitis media).

Q40. Answer: A — Increase in middle ear pressure can displace the graft

Increase middle ear pressure by nitrous oxide can displace the graft making nitrous oxide to be contraindicated for Tympanoplasties.

Q41. Answer: D — When surgeon is using 'inlay graft' (helps hold the graft)

If the surgeon is using 'inlay graft' then increase in middle ear pressure is beneficial in holding the graft.

Q42. Answer: B — Postoperative nausea and vomiting

The incidence of postoperative nausea and vomiting is very high in ear surgeries; therefore prophylaxis with antiemetics is must.

Q43. Answer: D — Nasal polyps, asthma, and sensitivity to aspirin/NSAIDs

In the preoperative evaluation always rule out the Samter triad (nasal polyps, asthma, and sensitivity to aspirin/NSAIDs).

Q44. Answer: C — To prevent aspiration of blood

Throat pack is must to prevent aspiration of blood.

Q45. Answer: C — Because they can become very restless in postoperative period after nasal packing if not trained

The patients must be counseled and trained in the preoperative period to take breathing from mouth otherwise they can become very restless in the postoperative period after nasal packing.

Q46. Answer: D — Because continuous monitoring of facial nerve function is needed

Continuous monitoring of facial nerve function may deter the anesthetist to use muscle relaxants.

Q47. Answer: C — All concerns related to obese patients especially difficult airway management

All concerns related to obese patients especially difficult airway management makes obstruction sleep apnea (OSA) surgeries (uvulopalatoplasties) to be very challenging to anesthetist.

Q48. Answer: D — Awake nasal intubation

Due to the limited mouth opening intubation mandates awake nasal intubation.

Q49. Answer: A — No, they are not 100% protective against laser fires

Laser resistant endotracheal tubes are not 100% protective against laser fires.

Q50. Answer: D — Laryngospasm

Laryngeal stimulation can induce laryngospasm.